

ResQGrid: Master Presentation & Defense Guide

Smart India Hackathon 2026 | Team BharatBytes (Team 16) | PS ID: SIH26191

Theme: Disaster Management | Category: Software | Complete Slide-by-Slide Script, Tech Explanation & Jury Defense

SLIDE 1: Title & Team Introduction

Slide Contents: Smart India Hackathon 2026 | Problem Statement ID: SIH26191 | Title: Intelligent Identification of Hazard-Based Red Zones, Carrying Capacity Assessment, and Immediate Relocation Needs for Vulnerable Habitations | Team ID: Team 16 | Team Name: BharatBytes

- **Why we use it:** Establishes domain authority, team identity, and problem statement focus right at the start of the pitch.
- **Software & Tools:** Official SIH Title Page Template with Team 16 credentials.
- **How it works:** The team leader opens the pitch clearly stating team name and core mission to save lives through proactive disaster intelligence.

Speaker Script (30 Seconds):

"Respected Jury and evaluators, a very good day to you. We are Team BharatBytes (Team ID 16). Today, we present **ResQGrid**, our AI-powered proactive relocation intelligence system designed for Problem Statement SIH26191: Intelligent Identification of Hazard-Based Red Zones, Carrying Capacity Assessment, and Immediate Relocation Needs for Vulnerable Habitations."

SLIDE 2: Solution Overview & Core Innovation (ResQGrid)

Slide Contents: Real-world Issue (Wayanad/Himachal landslides, unplanned evacuations), Rs 80k+ Cr annual loss, 30M+ displaced, 85% evacuation failures due to blind shelter allocation & road cutoffs. Solution: GeoAI Red-Zones + Multi-Resource Carrying Capacity + Capacity-Constrained Relocation + XAI (SHAP).

- **Why we use it:** Explains the urgent national disaster problem and clearly presents our three-pillar solution.
- **Software & Tools:** GeoAI raster fusion, SoVI demographic prioritization, Multi-Resource Capacity tracker, and Google OR-Tools optimization engine.
- **How it works:** 85% of evacuation failures occur due to blind shelter allocation and road cutoffs. ResQGrid predicts red zones before disaster strikes, monitors camp water/food/beds, and guarantees zero shelter overflow with safe routes.

Speaker Script (60 Seconds):

"In recent disasters like Wayanad and Himachal, 85% of evacuation failures occurred because authorities allocated people blindly to overcrowded shelters while convoys got trapped on severed roads. ResQGrid replaces chaotic evacuation with intelligence: it identifies dynamic red zones, tracks multi-resource shelter capacity (beds, water, food, sanitation), and uses mathematical optimization to safely evacuate every citizen with zero shelter overflow."

SLIDE 3: Technical Approach, Architecture & Implementation

Slide Contents: 5-Step Methodology (Data Ingestion, Dynamic GeoAI Red-Zone, SoVI Vulnerability, Carrying Capacity Modeling, OR-Tools Optimization), System Architecture Diagram, Linear Formulation, Technologies Used (FastAPI, React, PostGIS, OR-Tools, OSMNx).

- **Why we use it:** Proves technical feasibility and demonstrates that our team has built a working software system.
- **Software Stack:** FastAPI (Backend API), Google OR-Tools (MILP optimization), GeoPandas/Shapely (Geospatial engine), NetworkX/OSMNx (Dynamic road routing), React + Leaflet (Command Dashboard).
- **How it works:** Ingests DEM elevation and IMD rainfall -> flags dynamic Red Zones -> prioritizes vulnerable habitations (SoVI) -> calculates true multi-resource camp capacity -> dispatches rescue convoys with zero overflow in under 200ms.

Speaker Script (75 Seconds):

"Technically, our pipeline has 5 structured stages: GeoPandas ingests DEM terrain and IMD rainfall; our GeoAI engine calculates slope stability and rainfall thresholds to identify dynamic Red Zones; our SoVI module prioritizes vulnerable citizens; our Shelter Engine models capacity based on water, food, and beds; and Google OR-Tools executes a Mixed-Integer Linear Program to compute mathematically optimal, zero-overflow evacuation manifests in under 200 milliseconds."

SLIDE 4: Feasibility, Viability & Government Adoption

Slide Contents: Feasibility Rating (4.5/5), Infrastructure Reusability (ISRO open satellite feeds, state GIS), Scalable Pilot Deployment (Wayanad/Mandi), Explainable AI (SHAP for DM confidence), Government Savings & Zero Relief Wastage.

- **Why we use it:** Proves to judges that the system can be deployed immediately by State Disaster Management Authorities with near-zero extra infrastructure cost.
- **Software & Tools:** SHAP Explainability library, Open-source GIS connectors, Cloud PostgreSQL, and REST APIs.
- **How it works:** Reuses open government satellite and rainfall telemetry. Provides Explainable AI (SHAP) feature breakdowns so District Magistrates know exactly why an evacuation was ordered, ensuring legal compliance and public trust.

Speaker Script (45 Seconds):

"ResQGrid is 100% feasible and viable. It reuses existing government data feeds—ISRO Bhuvan, IMD weather feeds, and OpenStreetMap—requiring zero heavy hardware expenditure. Furthermore, we provide Explainable AI (SHAP): District Magistrates are not given a black box; they see exact percentage weights (e.g. 48% slope, 34% rainfall) behind every red zone, ensuring transparency, accountability, and zero relief wastage."

SLIDE 5: Impacts and Benefits

Slide Contents: 95% Reduced Evacuation Planning Latency (72 hrs to under 2 mins), Faster Targeted Response for vulnerable habitations, Zero Relief Material Stockouts, Real-Time Emergency Alerts. Pillars: Safety, Economic, Environmental, Technological.

- **Why we use it:** Quantifies the life-saving and financial benefits for disaster response agencies.
- **Software & Tools:** Benchmark analytics, automated convoy manifest generator, and real-time alert dispatch pipeline.
- **How it works:** Replaces slow manual spreadsheets with instant AI calculation, preventing camp resource exhaustion and saving lives during critical golden hours.

Speaker Script (40 Seconds):

"The impact of ResQGrid is transformative: It reduces evacuation decision time by 95%—from 72 hours of chaotic manual coordination down to under 2 minutes. It protects vulnerable citizens first, eliminates secondary humanitarian crises by preventing shelter stockouts, and saves district administrations crores in duplicate relief logistics."

SLIDE 6: Research, References & UI/UX Feature Preview

Slide Contents: References NDMA SOPs, ISRO NDEM, Cutter SoVI, Google OR-Tools, and Lundberg SHAP; previews core UI features: Dynamic Red-Zone Portal, Live Shelter Capacity Matrix, and Emergency Dispatch Console.

- **Why we use it:** Anchors the solution in verified scientific disaster management literature and proves operational software readiness.
- **Software & Tools:** NDMA Disaster SOPs, Sphere Humanitarian Standards, Google OR-Tools solver, and React/Leaflet UI.
- **How it works:** Combines established scientific research with a modern command center interface built for District Disaster Control Rooms.

Speaker Script (30 Seconds):

"Our solution is built on verified scientific standards—NDMA SOPs, Cutter's Social Vulnerability Index, and Sphere Humanitarian standards. As shown in our interface preview, ResQGrid delivers three core capabilities: Dynamic Hazard Red-Zoning, Live Multi-Resource Capacity Gauges, and 1-Click NDRF Convoy Dispatch Manifests. Thank you, and we are now open for questions!"

Top 5 Jury Questions & Winning Answers

Q1: How do you guarantee shelters will not get overcrowded during mass evacuations?

Answer: We enforce strict capacity constraints using Google OR-Tools Mixed Integer Linear Programming (MILP). The solver bounds total assigned evacuees to never exceed each shelter's effective capacity (minimum of beds, water, food, and sanitation). If a shelter reaches 100%, the solver automatically reroutes evacuees to the nearest safe secondary shelter.

Q2: What happens if a road or bridge gets washed away during an ongoing evacuation?

Answer: ResQGrid maintains a live OpenStreetMap road network via NetworkX. When a road is reported blocked or flooded, the software cuts that edge from the graph and OR-Tools instantly recalculates safe alternate detour routes in under 200ms.

Q3: Why shouldn't the District Magistrate just use Google Maps?

Answer: Google Maps only finds the shortest route for individual vehicles—it has zero knowledge of relief camp capacities, flood water levels, demographic vulnerability (elderly/disabled), or fleet coordination. ResQGrid is an end-to-end disaster logistics decision engine.

Q4: How do rescue teams receive orders if there is no internet in disaster zones?

Answer: ResQGrid generates 1-click printable PDF manifests at the District Command Center before dispatch, and supports compressed SMS alerts (160 characters) directly to Aapda Mitra and NDRF commanders over basic 2G cellular networks.

Q5: What makes your carrying capacity model different from existing systems?

Answer: Existing systems only count physical beds. ResQGrid uses a multi-resource degradation model compliant with Sphere Humanitarian Standards—tracking drinking water (15L/day), food calories (2100 kcal/day), sanitation ratios, and medical staff. If clean water drops, effective capacity drops dynamically to prevent secondary disease outbreaks.